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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Digital Student ID and Profile Management System

Submitted by

BAVATHARANI A

25BCS210

Under the Guidance of

Mr.Dharaneesh

Technical Trainer

Technical Department, KG College of Arts and Science

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Semester: **II**

DECLARATION

I hereby declare that the project entitled "**Digital Student ID and Profile Management System"** submitted in partial fulfillment of the requirements for the course **B.Sc.Computer Science** is a **record of my original work** carried out under the guidance of **Mr.Dharaneesh**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Bavatharani A

CERTIFICATE

This is to certify that the project entitled **"Digital Student ID and Profile Management System"** is a **bona fide work** carried out by **bavatharani A [1-'C'/2025-2026]**, **B.Sc.computer Science ,KG College of Arts & Science ,** in partial fulfillment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:Coimbatore
Date:10.04.2026

Submitted for the Viva-Voce Examination held on . **10/04/2026**

Guide:
(Technical Trainer, KGCAS)

Mr.Dharaneesh

Head of the Department
(Technical Head, KGCAS)

Dr.Boopalan

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ABSTRACT

The Digital Student ID and Profile Management System is a Java-based application designed to manage student information digitally. Traditional student record systems often rely on manual processes, which can lead to errors, data loss, and inefficiency.

This system provides a digital platform where student details such as ID, name, department, and profile information can be stored, updated, and retrieved easily. It uses Core Java concepts like classes, objects, and methods to create a simple and efficient system.

The main objective is to reduce paperwork, improve data accuracy, and provide quick access to student information.

1. INTRODUCTION

In educational institutions, managing student records is an important task. Manual record-keeping is time-consuming and prone to errors. With digital solutions, student data can be managed efficiently.
The Digital Student ID and Profile Management System helps store and manage student details electronically. It allows users to add, view, and update student records easily.

1.1 Objective:

- To develop a Java-based student management system
- To store student details digitally
- To reduce manual errors
- To provide easy access to student information
- To demonstrate Core Java concepts

1.2 Overview:

This system is a menu-driven Java application.
Users can:

- Enter student details
- View student profile
- Update information

The system ensures quick and accurate data handling

1.3 Features:

- Student ID Management
- Profile Storage
- Easy Data Retrieval
- Update Student Details
- User-Friendly Interface

2. PROGRAM ANALYSIS

2.1 Problem Definition:

Manual student record systems face issues like:

- Data loss
- Human errors
- Slow processing
- Difficulty in updating records

This system solves these problems using a digital approach.

2.2 Project Overview

The system allows:

- Adding student details
- Viewing student profile
- Managing student records

2.3 Scopes and Limitations :

scopes:

Schools and colleges Student data management

Limitations:

No database (if using basic Java) Limited to console application

3. SYSTEM DESIGN

3.1 Modules:

The Digital Student ID and Profile Management System is divided into different modules to perform specific tasks efficiently.

1.Student Entry Module:

This module is used to enter student details into the system.

The user can input important information such as Student ID, Name, and Department.

The data entered is stored in the system for future use.

2. Display Module:

This module is responsible for displaying the stored student details.
It retrieves the data and shows it in a clear and organized format, allowing users to easily view student information.

3. Update Module:

This module allows modification of existing student records.

Users can update details such as name or department if any changes are required. It

ensures that the student information remains accurate and up to date.

4. Exit Module:

This module is used to close the application safely.
When the user selects the exit option, the program terminates properly without any data loss.

4. METHODOLOGY

4.1 Problem Analysis:

In this phase, the problems of the existing manual system were identified. Manual record-keeping of student data leads to errors, data loss, and difficulty in accessing information. It is also time-consuming and inefficient. Therefore, a need for a digital system was recognized to improve accuracy and efficiency.

4.2 System Design:

The system was designed by dividing it into different modules such as student data entry, display, and update modules. The structure of the program was planned to ensure smooth flow and easy user interaction. A simple menu-driven design was used for better usability

4.3 Program Development:

The system was developed using Java programming language. Core concepts such as classes, objects, methods, loops, and conditional statements were used. The program was coded to accept user input, process data, and display the results correctly.

4.4 Testing and Validation:

Test Case	Expected Result
Enter student details	Details stored correctly
Display student profile	Correct information displayed
Update student details	Data updated successfully
Invalid input	Error handled properly

4.5 Result and Documentation:

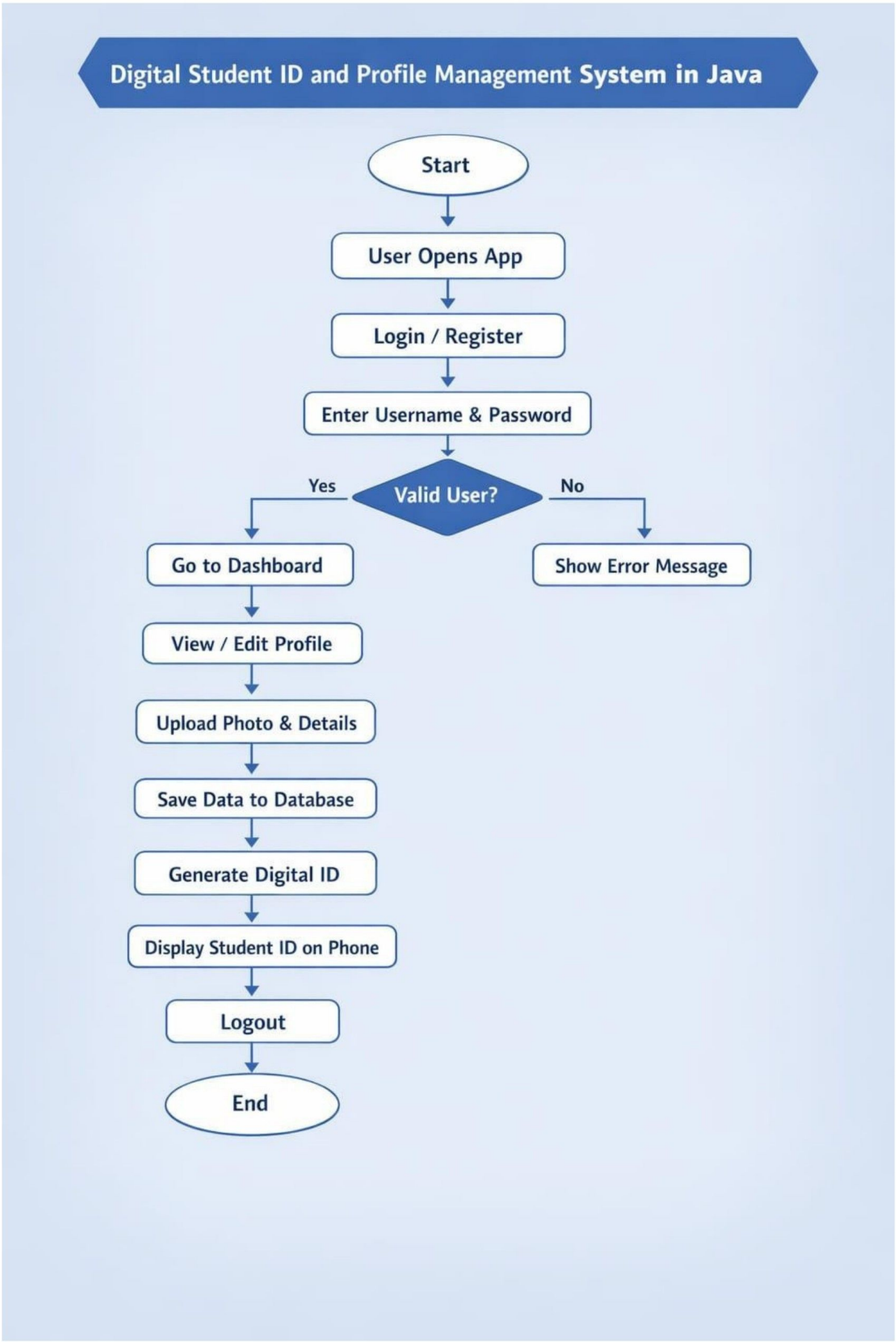
After testing, the system was found to work efficiently. It successfully stores and manages student data. The results were documented along with input and output samples, and the overall performance of the system was analyzed.

4.6Algorithm:

Algorithm for the **Digital Student ID and Profile Management System:**

1. Start
2. Enter student details
3. Store data
4. Display student profile
5. Update if needed
6. End

4.7 Flowchart:



5. SOURCE CODE

```
import java.util.Scanner;

class Student { int id;

String name; String department;


void getData() {
Scanner sc = new Scanner(System.in);

System.out.print("Enter Student ID: "); id = sc.nextInt();
sc.nextLine(); // clear buffer

System.out.print("Enter Student Name: "); name = sc.nextLine();

System.out.print("Enter Department: "); department =

sc.nextLine();
}


void displayData() {
System.out.println("\n----- DIGITAL STUDENT ID -----");

System.out.println("Student ID  : " + id);

System.out.println("Name : " + name);

System.out.println("Department  : " + department);
```

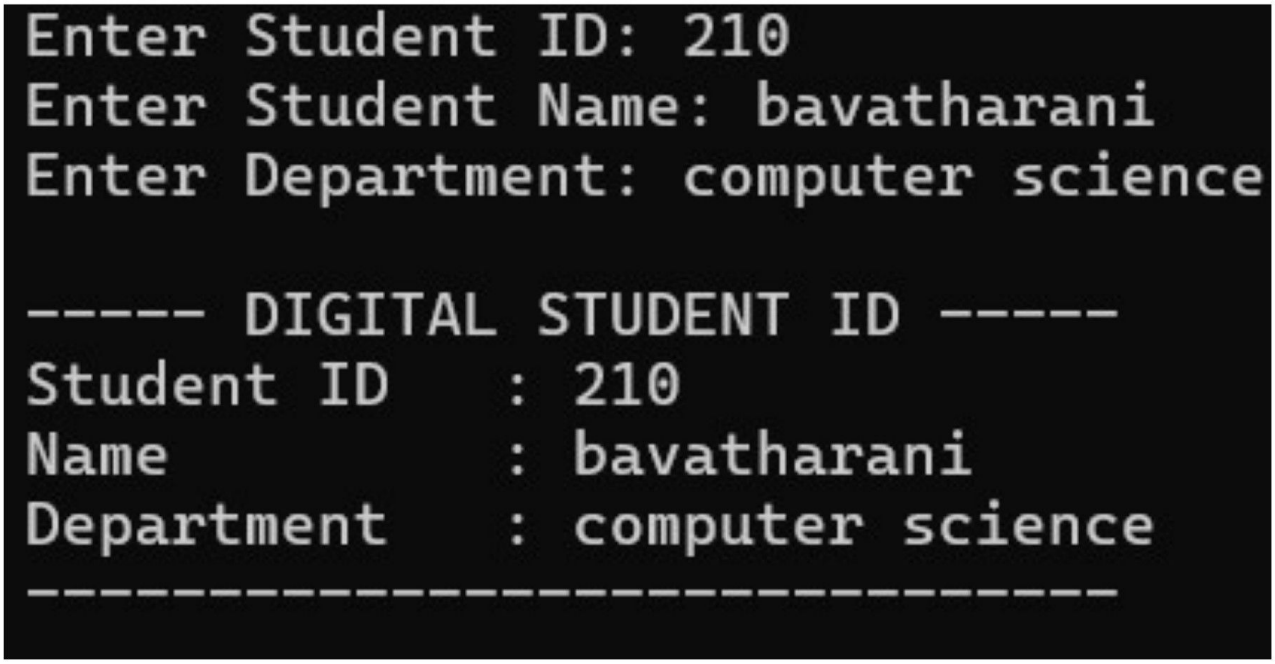
```
System.out.println("------------------------------");
}
}

public class StudentIDSystem {
public static void main(String[] args) { Student s = new
Student();

System.out.println("=== Digital Student ID System ===");

s.getData(); // input s.displayData(); // output
}
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 Discussion:

The Digital Student ID and Profile Management System was developed to manage student data in a digital format. The system allows users to enter, store, display, and update student information easily.

During development, Core Java concepts such as classes, objects, methods, and control structures were used effectively. The system was tested with different inputs, and it was observed that it performs all operations correctly.

Overall, the system reduces manual effort, minimizes errors, and provides a simple and efficient way to manage student records

7.2 Conclusion:

This project demonstrates how Java programming can be used to develop a simple real-world application. The system improves efficiency, accuracy, and data management compared to manual methods.

It serves as a basic model that can be further enhanced by adding features such as database connectivity, graphical user interface, and advanced search options in the future.

8. REFERENCES

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